AgileFlow AI

Proposal Presentation / Project Overview Summary

A Human-in-the-Loop Multi-Agent Framework for Agile SDLC Automation

Document Purpose	A compact project overview for knowledge passing, proposal slide preparation, new ChatGPT chats, and team onboarding.

Project ID	J26-SE-358
Project Name	AgileFlow AI - Human-in-the-Loop Multi-Agent Framework for Agile SDLC Automation
Project Type	Final-year Software Engineering research project
Core Theme	Human-supervised multi-agent automation for Agile SDLC activities
Primary Outcome	Structured Agile project environment, validated runnable prototype foundation, and traceability evidence

1. One-Minute Project Overview

AgileFlow AI is a human-in-the-loop multi-agent framework that supports small teams, start-ups, freelance groups, and student teams in moving from a raw client software idea to a structured Agile SDLC workflow and validated prototype foundation. The system connects requirement gathering, planning, design, pre-code testing, implementation, Git automation, testing, security validation, bug handling, and traceability through four specialized agents and a central Orchestrator.

The system does not claim to replace developers or produce a complete production-ready product. Its purpose is to create a controlled Agile project environment with clear artifacts, human approval gates, feedback loops, and evidence that the generated prototype follows the original client requirements.

2. Presentation Storyline

Step	Focus	Message for Presentation
1	Problem	Small teams struggle to convert raw client ideas into a connected Agile SDLC workflow.
2	Gap	Existing tools support separate tasks but do not fully connect requirements, design, code, testing, security, and approval.
3	Solution	AgileFlow AI uses specialized agents

		and an Orchestrator to automate and coordinate the SDLC workflow.
4	Human Control	Humans approve requirements, design, tests, code, and final validation reports.
5	Outcome	The result is a structured Agile environment, validated prototype foundation, and traceability evidence.

3. Problem Statement

Small software teams, start-ups, and student groups struggle to convert raw client ideas into a connected Agile SDLC workflow. Existing tools handle requirements, design, coding, testing, and security separately, causing unclear requirements, weak traceability, late defects, and limited human approval.

4. Proposed Solution

AgileFlow AI proposes a human-in-the-loop multi-agent framework that connects the Agile SDLC from client idea to validated prototype. It uses four agents and a central Orchestrator to automate requirements, planning, design, code generation, testing, security validation, and traceability while keeping humans involved through approval gates.

5. High-Level System Flow

Order	System Stage	Main Output
1	Client Input	Text, documents, web links, audio, video, or interview responses
2	Requirement Gathering and Analysis Agent	SRS, requirements, user stories, acceptance criteria, requirement IDs
3	Project Management and Design Agent	Backlog, sprint plan, UML, API plan, DB schema, UI flow, wireframe plan
4	Testing, Validation and Security Agent - Pre-Code	Test specifications, test skeletons, security test cases, requirement-to-test mapping
5	Implementation and Git Automation Agent	Generated prototype modules, repository structure, branches, commits, execution logs
6	Testing, Validation and Security Agent - Post-Code	Test report, bug report, security report, validation evidence
7	Final Output	Agile project environment, validated prototype foundation, reports, traceability matrix

6. Main Components

Component	Role in the System	Key Relationship
Requirement Gathering and Analysis Agent	Collects and structures client requirements from multiple input	Sends approved requirements to design, testing, implementation, and

	sources.	knowledge base.
Project Management and Design Agent	Converts approved requirements into Agile planning and design artifacts.	Provides design contracts to testing and implementation.
Testing, Validation and Security Agent	Creates pre-code test artifacts and performs post-code testing, regression, and security validation.	Guides implementation and reports failures to the Orchestrator.
Implementation and Git Automation Agent	Generates selected backend/frontend prototype modules and manages Git/workspace execution.	Uses approved requirements, designs, and tests; sends build output to validation.
Central SDLC Orchestrator	Controls workflow, approvals, handovers, retries, failure routing, and state.	Connects all agents and human approval gates.
Shared Knowledge Base	Stores approved artifacts, logs, reports, and traceability links.	Provides a common memory layer for all components.

7. Human-in-the-Loop Approval Gates

Gate	Approval Point	Purpose
H1	Requirement Approval	Human confirms that requirements and acceptance criteria are clear and valid.
H2	Design and Planning Approval	Human confirms that backlog, architecture, API, database, and UI plans are acceptable.
H3	Pre-Code Test Approval	Human confirms that test specifications match requirements and design expectations.
H4	Implementation Approval	Human reviews generated prototype modules, Git metadata, and execution logs.
H5	Final Validation Approval	Human reviews test results, security findings, bug reports, and final traceability evidence.

8. Research Gap and Novelty

Existing tools are useful for isolated tasks such as Jira planning, UI design, code generation, testing, or security scanning. The research gap is the lack of one controlled workflow that connects these stages with shared state, human approval, correction loops, and traceability.

The novelty of AgileFlow AI is not a single prompt or ordinary tool integration. The contribution is the coordinated human-in-the-loop multi-agent SDLC workflow, where requirement, design, implementation, testing, security, orchestration, and traceability are connected through structured artifacts and approval gates.

9. Target Users and Impact

Target Users	Expected Impact
Small software development teams	Reduce rework and improve SDLC structure.

Start-ups and freelance teams	Speed up prototype planning and validation.
Student project groups	Follow a clearer Agile workflow with evidence.
Supervisors, lecturers, and evaluators	Review requirements, design, code, tests, security, and traceability more easily.

- Reduces project rework and improves requirement clarity.
- Improves traceability from client idea to validated prototype.
- Supports earlier bug detection and security validation.
- Keeps humans in control through approval gates.

10. Scope Clarification

In Scope	Out of Scope / Not Claimed
Generating structured SDLC artifacts and prototype foundation	Replacing the full software development team
Human-supervised workflow with approval gates and traceability	Automatically delivering a complete production-ready system
Controlled sample domains such as book, library, tuition, appointment, or e-commerce systems	Guaranteeing perfect requirements, code, tests, or security without human validation

11. Suggested Proposal Slide Structure

Slide Area	Content	Purpose
Opening	Project title and team details	Introduce AgileFlow AI clearly.
Problem	Small teams, missing roles, disconnected SDLC	Show why the research is needed.
Research Gap	Existing tools work separately	Show academic gap.
Solution	Four agents, Orchestrator, Knowledge Base, approval gates	Explain proposed system.
Architecture	High-level system architecture and relationship diagram	Show how components connect.
Component Slides	Each member explains their agent	Show individual contribution.
Human-in-the-Loop	H1 to H5 approval gates	Show responsible AI and human control.
Traceability	Requirement-to-design-to-code-to-test links	Show research value.
Evaluation	Accuracy, coverage, build success, test pass rate, security findings, user review	Show measurable validation.
Commercialization and Timeline	Target market, value, revenue model, Gantt/WBS	Show feasibility and planning.

12. Evaluation Ideas

Area	Example Metric
Requirements	Requirement extraction accuracy, ambiguity detection,

	human approval score
Design and Planning	Requirement-to-design coverage, API/schema validity, backlog completeness
Implementation	Build success rate, Docker run success, self-fix success rate, traceability to code
Testing	Test coverage, test pass rate, failed-test classification accuracy
Security	Number/severity of findings, false-positive review, re-scan success
Whole System	End-to-end workflow completion, traceability matrix completeness, user satisfaction

13. Final Presentation Message

AgileFlow AI helps transform a raw client software idea into a connected Agile workflow and validated prototype foundation - with automation support, human approval, correction loops, and end-to-end traceability.